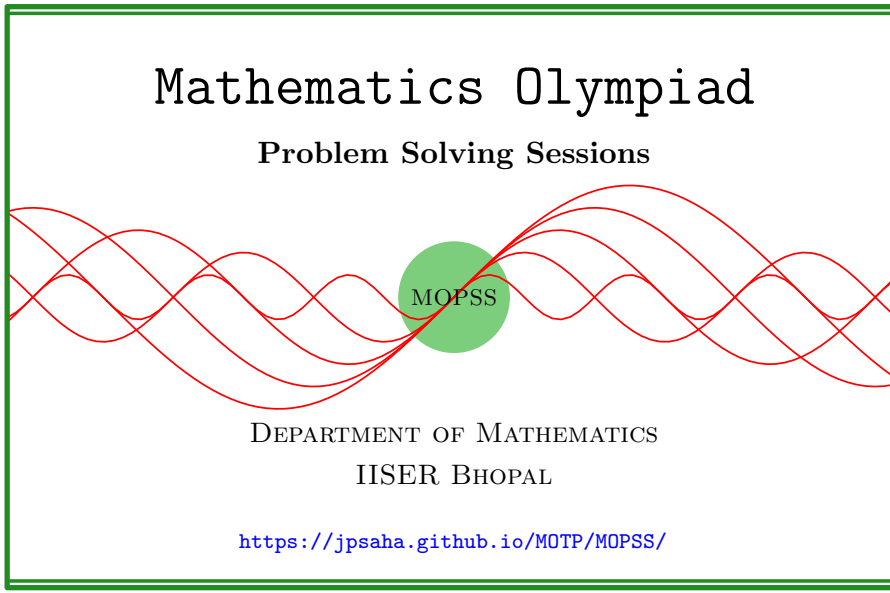
Optimization problems



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (Princeton University Mathematics Competition 2009 Combinatorics A6,) . . . 3
2 Exercise (Tournament of Towns 2014 Spring Junior A Level P2, , by A. V. Shapovalov) . . . 3
3 Exercise (Tournament of Towns 2014 Spring Senior A Level P3, by I. I. Bogdanov) . . . 3
4 Exercise (Junior Balkan MO 2015 P4,) . . . 4
5 Exercise (British Mathematical Olympiad Round 1 2015/16 P4) 4
6 Exercise (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P2, , proposed by M. A. Evdokimov) . . . 5
7 Exercise (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P5, , proposed by P. A. Borodin) . . . 7
8 Exercise (Benelux Mathematical Olympiad 2020 P2,) . 7
9 Exercise (Swiss Mathematical Olympiad 2023 Final Round Day 2 P8,) . . . 9
10 Exercise (Kosovo National Olympiad 2024 Grade 8 P3,) 9
11 Exercise (Junior Balkan MO 2019 P4,) . . . 10
12 Exercise (Argentina National Olympiad 2020 Level 2 P2,) 10
13 Exercise (Brazil National Olympiad 2022 Level 3 P6,) . 11
14 Exercise (Junior Balkan MO Shortlist 2022 C1,) . . . 11
15 Exercise (Junior Balkan MO Shortlist 2022 C3,) . . . 11
16 Exercise (Junior Balkan MO Shortlist 2023 C1,) . . . 12
17 Exercise (Canadian Junior Mathematical Olympiad 2023 P3, , cf. CMO 2023 P1) . . . 12
18 Exercise (Rioplatense Mathematical Olympiad 2025 Level 3 Grade 12 P2,) . . . 12
19 Exercise (All-Russian Mathematical Olympiad 2025 Grade 10 Day 2 P5, , also 11.5, proposed by I. I. Bogdanov) . 13
20 Exercise (RMO 2016g P2,) . . . 14
21 Exercise (RMO 2017a P4,) . . . 14
22 Exercise (RMO 2016c P4,) . . . 19
23 Exercise (RMO 2018b P4,) . . . 23
24 Exercise (INMO 2025 P5, , proposed by Pranjal Srivastava and Rohan Goyal) . . . 24
25 Exercise (Moscow Mathematical Olympiad 2025 Grade 9 P4, proposed by A. Zaslavsky) . . . 28
26 Exercise (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P6, , proposed by S. L. Berlov) . . . 28

The content posted here and at this blog by Evan Chen are quite useful.

§1 Optimization problems

Exercise 1 (Princeton University Mathematics Competition 2009 Combinatorics A6,). We have a 6×6 square, partitioned into 36 unit squares. We select some of these unit squares and draw some of their diagonals, subject to the condition that no two diagonals we draw have any common points. What is the maximal number of diagonals that we can draw?

> **Walkthrough** —
> **(a)**

Solution 1. to write ■

> **Remark.** See also this on AoPS.

Exercise 2 (Tournament of Towns 2014 Spring Junior A Level P2,, by A. V. Shapovalov). Peter marks several cells on a 5×5 board. Basil wins if he can cover all marked cells with three-cell corners. The corners must be inside the board and not overlap. What is the least number of cells Peter should mark to prevent Basil from winning? (Cells of the corners must coincide with the cells of the board).

> **Walkthrough** —
> **(a)**

Solution 2. to write ■

Exercise 3 (Tournament of Towns 2014 Spring Senior A Level P3, by I. I. Bogdanov). The King called two wizards. He ordered First Wizard to write down 100 positive real numbers (not necessarily distinct) on cards without revealing them to Second Wizard. Second Wizard must correctly determine all these numbers, otherwise both wizards will lose their heads. First Wizard is allowed to provide Second Wizard with a list of distinct numbers, each of which is either one of the numbers on the cards or a sum of some of these numbers. He is not allowed to tell which numbers are on the cards and which numbers are their sums. Finally the King tears as many hairs from each wizard's beard as the number of numbers in the list given to Second Wizard. What is the minimal number of hairs each wizard should lose to stay alive?

Walkthrough —

(a)

Solution 3. to write. ■

Exercise 4 (Junior Balkan MO 2015 P4,). An L-shape is one of the following four pieces, each consisting of three unit squares:

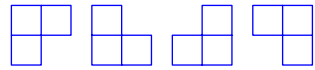


Figure 1: Junior Balkan MO 2015 P4, Exercise 4

A 5×5 board, consisting of 25 unit squares, a positive integer $k \leq 25$ and an unlimited supply of L-shapes are given. Two players A and B, play the following game: starting with A they play alternatively mark a previously unmarked unit square until they marked a total of k unit squares.

A placement of L-shapes on unmarked unit squares is called *good* if the L-shapes do not overlap and each of them covers exactly three unmarked unit squares of the board.

B wins if every good placement of L-shapes leaves uncovered at least three unmarked unit squares. Determine the minimum value of k for which B has a winning strategy.

Exercise 5 (British Mathematical Olympiad Round 1 2015/16 P4). James has a red jar, a blue jar and a pile of 100 pebbles. Initially both jars are empty. A move consists of moving a pebble from the pile into one of the jars or returning a pebble from one of the jars to the pile. The numbers of pebbles in the red and blue jars determine the state of the game. The following conditions must be satisfied:

(a) The red jar may never contain fewer pebbles than the blue jar;

(b) The game may never be returned to a previous state.

What is the maximum number of moves that James can make?

Walkthrough — Suppose there are n pebbles at the beginning, where n is even.

(a) Note that the state can be thought as a pair (r, b) where r and b denote the number of pebbles in the red and the blue jar respectively, at some instance.

(b) Observe that $r \geq b$ and $r + b \leq n$.

(c) This shows that (r, b) lies in a specific triangular region on the plane.

(d) Conclude that the maximum number of moves that James can make is at

most the total number of lattice points in this triangular region, which is equal to

$$T := \sum_{r=0}^{n} \sum_{b=0}^{\min\{r,n-r\}} 1 = 1 + 3 + 5 + \cdots + (n+1) = (n/2+1)^2.$$

(e) Show that there is a path in this triangular region beginning at the origin, and traversing through at least

$$T - \frac{n}{2}$$

points in this region. Does Figs. 2 and 3 help?

(f) Observe that each move changes the parity of the sum of the coordinates.

(g) Show that after make any number of moves, the number of states with an even sum of coordinates is at most one more than the number of states with an odd sum of coordinates.

(h) Prove that the number of points in the triangular region with an even sum of coordinates is precisely $n/2+1$ more than those with an odd sum of coordinates. See Figs. 4 and 5.

(i) If James makes an odd number of moves, say $2k-1$ moves, then show that the number of points on the path formed is

$$2k \leq 2O = T - n/2 - 1.$$

(j) Suppose James makes an even number of moves, say, $2k$ moves. Conclude that the number of points on the path formed is

$$k + k + 1 \leq 2O + 1 = T - n/2 - 1 + 1 = T - n/2.$$

(k) Show that the maximum number of moves that James can make is

$$T - n/2 - 1 = (n/2+1)^2 - n/2 - 1 = \frac{n^2+2n}{4}.$$

Exercise 6 (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P2, , proposed by M. A. Evdokimov)**.** A number of dominoes were placed without overlapping on a 2018×2018 square board, each covering exactly two squares. It turned out that no two dominoes share a common whole side, that is, no two dominoes form either a 2×2 square or a 4×1 rectangle. Is it possible to cover more than 99% of all the squares on the board?

Walkthrough —

(a)

Solution 4. to write ■

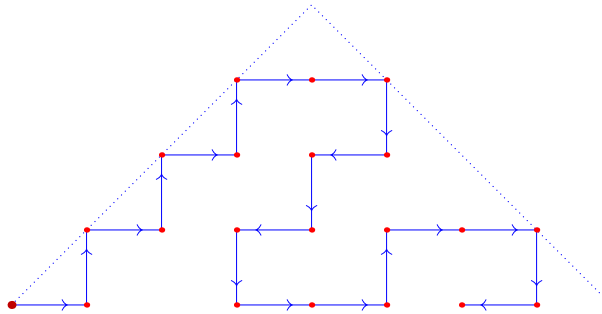


Figure 2: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 5

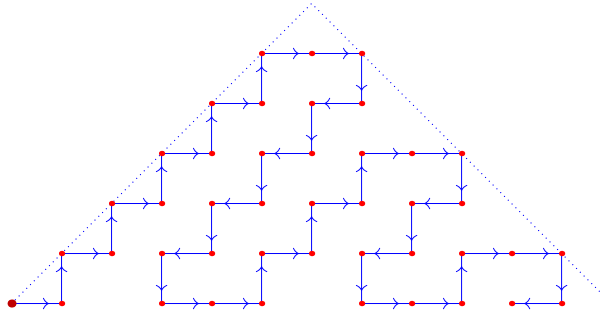


Figure 3: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 5

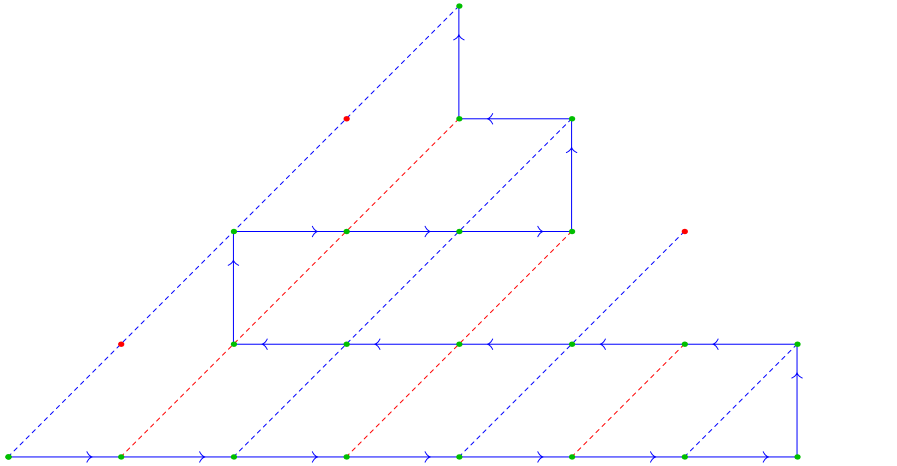


Figure 4: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 5

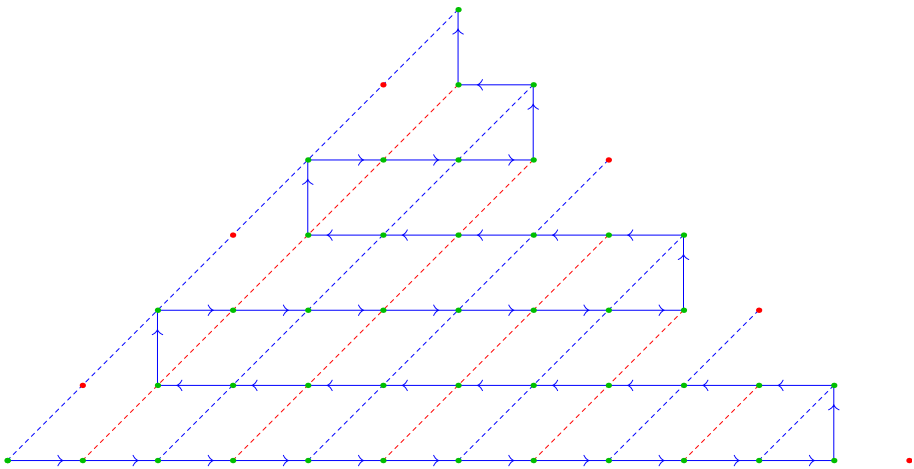


Figure 5: British Mathematical Olympiad Round 1 2015/16 P4, Exercise 5

Exercise 7 (Moscow Mathematical Olympiad 2018 Grade 11 Day 2 P5, , proposed by P. A. Borodin)**.** Zhenya painted a spherical egg sequentially using five different dyes, dipping it into a glass containing the current dye in such a way that exactly half of the egg's surface area (a hemisphere) was colored. As a result, the egg became completely colored. Prove that one of the dyes was superfluous, that is, if Zhenya had omitted that particular dye but had dipped the egg into the remaining dyes in the exact same manner, the egg would still have ended up completely colored.

Walkthrough —

(a)

Solution 5. to write ■

Remark. We refer to Carathéodory's theorem on convex hulls.

Exercise 8 (Benelux Mathematical Olympiad 2020 P2,)**.** Let N be a positive integer. A collection of $4N^2$ unit tiles with two segments drawn on them as shown in Fig. 6 is assembled into a $2N \times 2N$ board. Tiles can be rotated.

The segments on the tiles define paths on the board. Determine the least possible number and the largest possible number of such paths.

For example, there are 9 *paths on the* 4×4 *board shown in Fig. 7.*

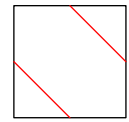


Figure 6: BxMO 2020 P2, Exercise 8

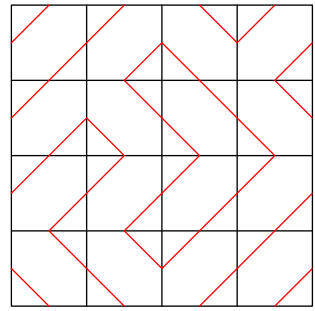


Figure 7: BxMO 2020 P2, Exercise 8

Walkthrough —

(a)

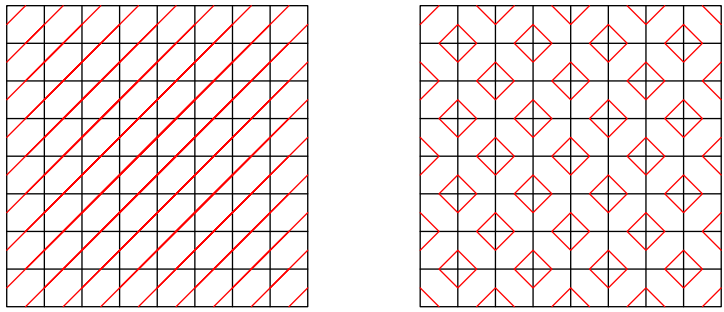
Solution 6. Consider a given tiling of a $2N \times 2N$ board using tiles of the given type. Note that there are two types of paths. Some path are between two points lying on the boundary of the square, and the other paths are contained in the interior of the square, and form cycles. Let $\mathcal{B}, \mathcal{C}$ denote the collection of paths of the first and second kind respectively. Observe that among the mid-points of the sides of these $4N^2$ tiles, there are precisely $4 \times 2N = 8N$ mid-points which lie on the boundary of the square. Since each of the paths, which are not cycles, have precisely two such points as end-points, and no two paths of these kind have an end-point in common, it follows that there are precisely $4N$ paths, which are not cycles.

Observe that no two paths have a segment in common. Note that the remaining paths, which are contained in the interior of the square, form cycles. Each of these cycles contains at least four segments. Hence, these cycles together contain at least $4|\mathcal{C}|$ segments. Also note that a path contains at least two segments, unless it is formed by single segment, and note that there are at least four such paths. Let s denote the number of paths consisting of a single segment. Since there are a total of $4N^2$ tiles, each contributing two segments, we have $8N^2$ segments in total. This shows that

$$8N^2 \geq s + 2(|\mathcal{B}| - s) + 4|\mathcal{C}|,$$

which yields

$$8N^2 + 8N \geq 4(|\mathcal{B}| + |\mathcal{C}|) - s.$$



(a) (b)

Figure 8: BxMO 2020 P2, Exercise 8

It follows that

$$|\mathcal{B}| + |\mathcal{C}| \leq 2N^2 + 2N + \frac{s}{4} \leq 2N^2 + 2N + 1.$$

This proves that there are exactly $4N$ paths which are not cycles, and there are at most $2N^2 + 2N + 1$ paths in total.

The tiling, as illustrated in Figure 8a in the case $N = 4$, shows an example of a configuration where the number of paths is equal to $4N$. The tiling, as illustrated in Figure 8b in the case $N = 4$, shows an example of a configuration where the number of paths is equal to $2N^2 + 2N + 1$.

Hence, the least possible number and the maximum possible number of paths are $4N$ and $2N^2 + 2N + 1$ respectively. ■

Exercise 9 (Swiss Mathematical Olympiad 2023 Final Round Day 2 P8,). Let n be a positive integer. We start with n piles of pebbles, each initially containing a single pebble. One can perform moves of the following form: choose two piles, take an equal number of pebbles from each pile and form a new pile out of these pebbles. Find (in terms of n) the smallest number of nonempty piles that one can obtain by performing a finite sequence of moves of this form.

> **Walkthrough** —
>
> **(a)**

Solution 7. to write ■

Exercise 10 (Kosovo National Olympiad 2024 Grade 8 P3,). The numbers $1, 2, 3, \ldots, 20$ are placed in any order in a circle. A number is considered to be *high* if it is larger than the two numbers near it. What is the greatest possible number of high numbers?

Walkthrough —

(a) Prove that there are at most 10 high numbers in any arrangement of $1, 2, \ldots, 20$ on a circle.

(b) Find an arrangement of $1, 2, \ldots, 20$ on a circle having 10 high numbers.

Solution 8. Suppose we are given an arrangement of $1, 2, \ldots, 20$ on a circle. Let k denote the number of high numbers in this arrangement. Since 20 is a high number, we have $k \geq 1$. Let $h_1, h_2, \ldots, h_k$ denote the high numbers in this arrangement. Assume that further that these high numbers are arranged in the order $h_1, h_2, \ldots, h_k$ on the circle. Let us consider case that $k \geq 2$. Observe that two high numbers cannot be adjacent to each other. Therefore, there must be at least one non-high number between any two high numbers. Therefore, there are at least k non-high numbers in this arrangement. Since there are 20 numbers in total, there are at most $20 - k$ non-high numbers. Therefore, we have $k \leq 20 - k$, which implies that $k \leq 10$ under the assumption that $k \geq 2$. This proves that $k \leq 10$, that is, there are at most 10 high numbers in any arrangement of $1, 2, \ldots, 20$ on a circle.

Consider the arrangement

$$1, 11, 2, 12, 3, 13, 4, 14, 5, 15, 6, 16, 7, 17, 8, 18, 9, 19, 10, 20$$

on a circle. In this arrangement, the high numbers are 11, 12, 13, 14, 15, 16, 17, 18, 19, 20. Therefore, there are 10 high numbers in this arrangement.

Therefore, the greatest possible number of high numbers is equal to 10. ■

Exercise 11 (Junior Balkan MO 2019 P4,). A 5×100 table is divided into 500 unit square cells, where n of them are coloured black and the rest are coloured white. Two unit square cells are called adjacent if they share a common side. Each of the unit square cells has at most two adjacent black unit square cells. Find the largest possible value of n.

Walkthrough —

(a)

Solution 9. to write ■

Exercise 12 (Argentina National Olympiad 2020 Level 2 P2,). Let n be a positive integer. There are n colors available. Each of the integers from 1 to 1000 must be painted with one of the n colors such that any two different numbers, if one divides the other, are painted in different colors. Determine the smallest value of n for which this is possible.

Walkthrough —

(a)

Solution 10.

■

Exercise 13 (Brazil National Olympiad 2022 Level 3 P6,). Some cells of a 10×10 chessboard are colored blue. A set of six cells is called *gremista* when the cells are the intersection of three rows and two columns, or two rows and three columns, and are painted blue. Determine the greatest value of n for which it is possible to color n chessboard cells blue such that there is not a gremista set.

Walkthrough —

(a)

Solution 11.

■

Remark. This problem is equivalent to finding the maximum possible number of edges in a (possibly directed) graph on 10 vertices which is $K_{3,2}$-free.

Exercise 14 (Junior Balkan MO Shortlist 2022 C1,). Anna and Bob, with Anna starting first, alternately color the integers of the set $S = \{1, 2, \ldots, 2022\}$ red or blue. At their turn each one can color any uncolored number of S they wish with any color they wish. The game ends when all numbers of S get colored. Let N be the number of pairs (a, b), where a and b are elements of S, such that a, b have the same color, and $b - a = 3$. Anna wishes to maximize N. What is the maximum value of N that she can achieve regardless of how Bob plays?

Walkthrough —

(a)

Solution 12.

■

Exercise 15 (Junior Balkan MO Shortlist 2022 C3,). There are 200 boxes on the table. In the beginning, each of the boxes contains a positive integer (the integers are not necessarily distinct). Every minute, Alice makes

one move. A move consists of the following. First, she picks a box X which contains a number c such that $c = a + b$ for some numbers a and b which are contained in some other boxes. Then she picks a positive integer $k > 1$. Finally, she removes c from X and replaces it with kc. If she cannot make any moves, she stops. Prove that no matter how Alice makes her moves, she won't be able to make infinitely many moves.

> **Walkthrough** —
>
> **(a)**

Solution 13.

■

Exercise 16 (Junior Balkan MO Shortlist 2023 C1,). Given is a square board with dimensions 2023×2023, in which each unit cell is colored blue or red. There are exactly 1012 rows in which the majority of cells are blue, and exactly 1012 columns in which the majority of cells are red. What is the maximal possible side length of the largest monochromatic square?

> **Walkthrough** —
>
> **(a)**

Solution 14.

■

Exercise 17 (Canadian Junior Mathematical Olympiad 2023 P3, , cf. CMO 2023 P1). William is thinking of an integer between 1 and 50, inclusive. Victor can choose a positive integer m and ask William: "does m divide your number?", to which William must answer truthfully. Victor continues asking these questions until he determines William's number. What is the minimum number of questions that Victor needs to guarantee this?

> **Walkthrough** —
>
> **(a)**

Solution 15.

■

Exercise 18 (Rioplatense Mathematical Olympiad 2025 Level 3 Grade 12 P2,). Let $n \geq 2$ be an integer. There are n numbers, each one starts at 0. In

each operation, one chooses two numbers, the smallest increases by 1 and the biggest becomes 0 (if they are equal, one increases 1 and the other becomes 0). This process can be repeated indefinitely. Find, in function of n, the biggest number that can appear in the sequence.

Exercise 19 (All-Russian Mathematical Olympiad 2025 Grade 10 Day 2 P5, , also 11.5, proposed by I. I. Bogdanov)**.** Let n be a natural number. The numbers $1, 2, \ldots, n$ are written in a row in some order. For each pair of adjacent numbers, their greatest common divisor is calculated and written on a sheet. What is the maximum possible number of distinct values among the $n-1$ integers obtained?

Walkthrough —

(a)

Solution 16. In the following, we assume that $n \neq 1$. Using division algorithm, for any two distinct positive integers a and b, it follows that

$$\gcd(a,b) \leq \begin{cases} a \leq \frac{b}{2} & \text{if } a \text{ divides } b, \\ \frac{a}{2} & \text{if } a \text{ does not divide } b, \end{cases}$$

and hence $\gcd(a,b) \leq \frac{1}{2}\max(a,b)$. This implies that the GCD of any two adjacent numbers in the row is at most $n/2$. Therefore, the maximum possible number of distinct GCDs obtained, is at most $\lfloor n/2 \rfloor$.

We will show that this upper bound can be achieved by the following arrangement of the numbers $1, 2, \ldots, n$. For each odd positive integer $k \leq n$, write the integers between 1 and n in a block $\mathcal{B}_k$, having k as its largest odd divisor, in an increasing order, that is, the elements of $\mathcal{B}_k$ are put in the order

$$k, 2k, 4k, \ldots, 2^{e_k}k,$$

where e_k is the largest non-negative integer such that $2^{e_k}k \leq n$. Next, we put the blocks $\mathcal{B}_k$ in a row as k increases across the odd positive integers less than or equal to n.

Claim — Let $1 \leq m \leq n/2$ be a positive integer. Then m is equal to the GCD of some pair of adjacent numbers in the above arrangement.

Proof of the Claim. Since m is at most $n/2$, it follows that $2m$ is at most n. If k denotes the largest odd divisor of m, then k is less than or equal to m and hence $k \leq n$. Hence, in the above arrangement, the block $\mathcal{B}_k$ has appeared, and the numbers m and $2m$ are adjacent in this block. Note that the GCD of m and $2m$ is m. This proves the claim. □

Therefore, the GCDs of the adjacent numbers in the above arrangement include all the integers $1, 2, \ldots, \lfloor n/2 \rfloor$. Since the number of distinct GCDs obtained is at most $\lfloor n/2 \rfloor$, it follows that the maximum possible number of distinct GCDs obtained is $\lfloor n/2 \rfloor$. ■

Remark. One may also observe that if ℓ denotes the largest GCD obtained from the adjacent numbers in the row, then it is the gcd of two distinct integers between 1 and n, and the larger of these two integers is at least 2ℓ, implying that $\ell \leq n/2$.

Exercise 20 (RMO 2016g P2,). On a stormy night ten guests came to dinner party and left their shoes outside the room in order to keep the carpet clean. After the dinner there was a blackout, and the guests leaving one by one, put on at random, any pair of shoes big enough for their feet. (Each pair of shoes stays together). Any guest who could not find a pair big enough spent the night there. What is the largest number of guests who might have had to spend the night there?

Walkthrough — What happens if a person having shoe of the smallest size wears a shoe of the largest size, and next, if a person having shoe of the second smallest size wears a shoe of the second largest size, and it continues?

Solution 17. If a person having shoe of the k-th smallest size wears a shoe of the k-th largest size for $k = 1, 2, 3, 4, 5$, and if the sizes of the shoes of the ten persons are distinct, then none of the remaining five persons will not find a big enough shoe.

We claim that if k persons have left for some $1 \leq k \leq 4$, then one of the remaining $10 - k$ persons will find a big enough shoe. Let us denote the sizes of the shoes of the guests by $s_1, s_2, \ldots, s_{10}$. If the claim is false, then for some $1 \leq k \leq 4$, there are $10 - k$ guests whose shoes have sizes larger than some $10 - k$ numbers among $s_1, \ldots, s_{10}$. So, some $10 - k$ numbers among $s_1, \ldots, s_{10}$ are larger than some $10 - k$ numbers among $s_1, \ldots, s_{10}$. It follows that

$$10 - k + 10 - k \leq 10,$$

which shows that $k \geq 5$. This contradicts the bound $k \leq 4$. This proves the claim. Consequently, at most five guests may need to spend the night there.

We conclude that the largest number of the guests who might need to spend the night there is five. ■

Exercise 21 (RMO 2017a P4,). Consider n^2 unit squares in the xy-plane centred at the point (i, j) with integer coordinates for $1 \leq i \leq n, 1 \leq j \leq n$. It is required to colour each unit square in such a way that whenever $1 \leq i < j \leq n$ and $1 \leq k < \ell \leq n$, the three squares with centres at (i, k), (j, k), (j, ℓ) have distinct colours. What is the least possible number of colours needed?

Remark. For such problems, it is often useful to first work out a special case.

Walkthrough —

(a) First, work out a simple case in order to gain insight for the general case.

(b) One may consider the squares below a suitable diagonal.

(c) Extend to the general case!

First, let's work on it. Let us begin with the case $n = 8$. First, let us try to color the unit squares with as few colors as we can. This may provide some insight for the least number of colors required (for the case $n = 8$ and also possibly for the general case).

Note that any two unit squares at the bottom row of Fig. 9a have pairwise distinct colors. Let's apply the colors $1, \ldots, 8$ to these squares as in Fig. 9b. Note that in the second last row, all the unit squares except the first one, have colors different from those of the unit squares of the bottom row. Moreover, these seven unit squares have distinct colors. Let's apply the colors $3, \ldots, 9$ to these squares as in Fig. 9c. Similarly, in the third last row, the $8 - 3 + 1$ unit squares lying to the right, have colors different from those of the unit squares have been colored so far. Moreover, these $8 - 3 + 1$ unit squares have distinct colors. Let's apply another set of $8 - 3 + 1$ new colors (for example, $5, \ldots, 10$) to these squares as in Fig. 9d. One may continue this process to yield the coloring as in Fig. 9e.

In the remaining square in the second last row, we may use a color which has been used in that row, for instance, the color 2, as in Fig. 10a. In the third last row, the colors $3, 4$ may be used as in Fig. 10b. In the fourth last row, the colors smaller than 7 may be used as in Fig. 10c. One may continue this process to yield the coloring as in Fig. 10d. Note that the coloring as in Fig. 10d **does satisfy** the required condition. ♣

Remark. The **preceding argument alone** does not guarantee that the least possible number of colors have been used in Fig. 10d.

Skip this part for the first reading —

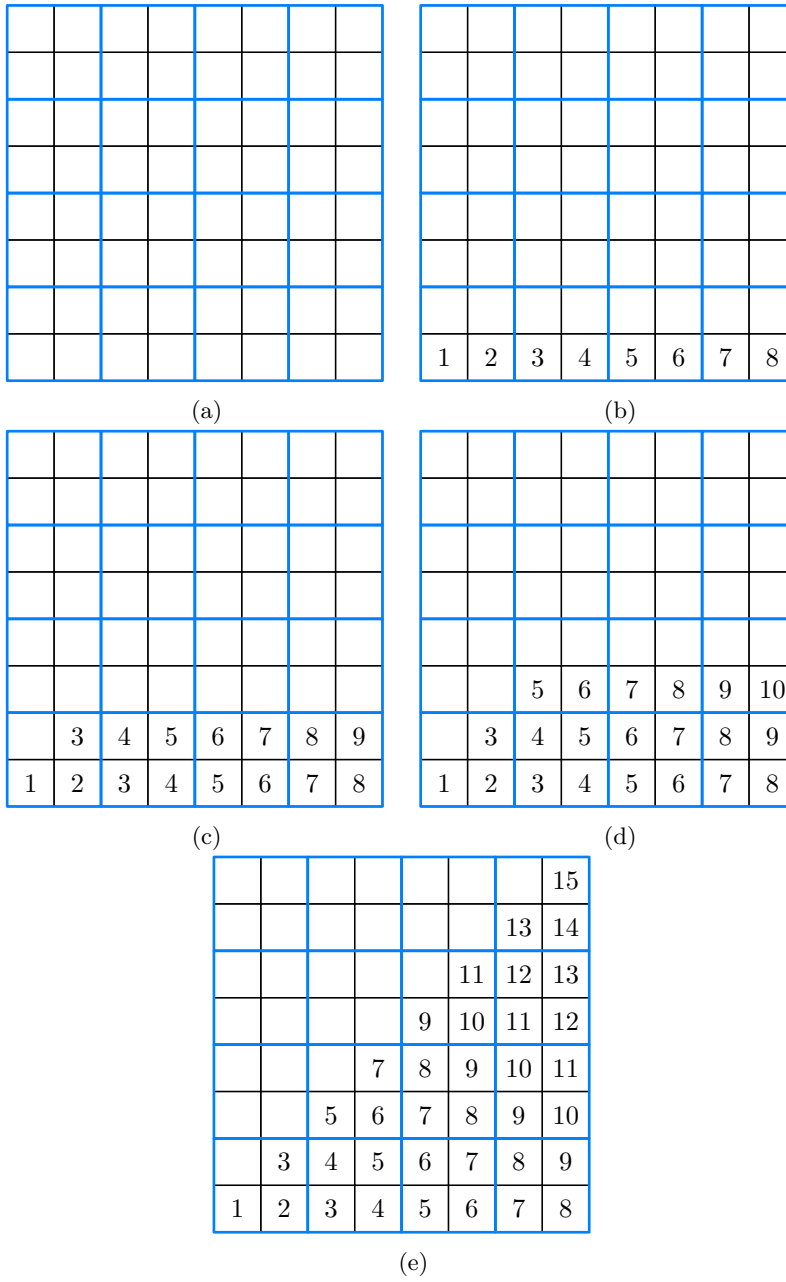


Figure 9: India RMO 2017a P4, Exercise 21

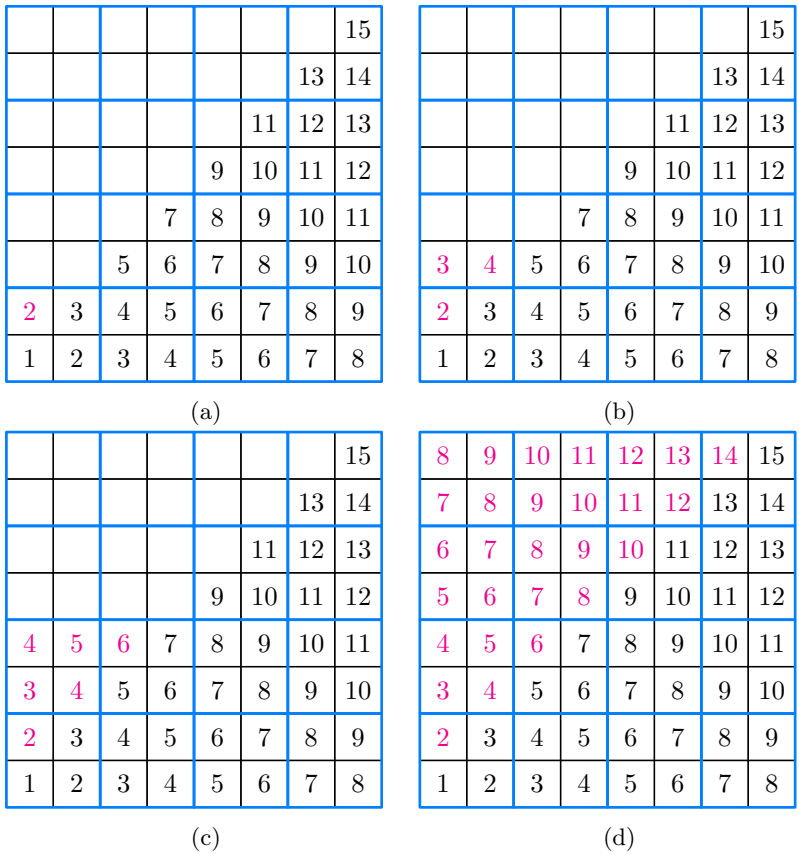


Figure 10: India RMO 2017a P4, Exercise 21

Remark. The **preceding argument alone** does not guarantee that the least possible number of colors have been used in Fig. 10d. It could have been the case that in our very first attempt of working out a special case, we might have proceeded in way which would have used more colors than what is required. For instance, consider the following approach.

In fact, that is what I (the author) did in a prior attempt, and then I could not prove that the number of colors that one may require to color the squares satisfying the required condition is at least as large as the number of colors used in the approach below. This led to think again, which yield the argument above.

¶ A prior attempt that used more colors than the minimum number of colors required. Let us begin with the case $n = 8$. First, let us try to color the unit squares with as few colors as we can. This may provide some insight for the least number of colors required (for the case $n = 8$ and also possibly for the general case).

Note that any two unit squares at the bottom row of Fig. 11a have pairwise distinct colors. Let us apply the colors $1, \ldots, 8$ to these squares as in Fig. 11b. Note that in the second last row, all the unit squares except the first one, have colors different from those of the unit squares of the bottom row. Moreover, these seven unit squares have distinct colors. Let us apply the colors $9, \ldots, 15$ to these squares as in Fig. 11c. Similarly, in the third last row, the $8-3+1$ unit squares lying to the right, have colors different from those of the unit squares have been colored so far. Moreover, these $8-3+1$ unit squares have distinct colors. Let us apply another set of $8-3+1$ new colors (for example, $16, \ldots, 21$) to these squares as in Fig. 11d. One may continue this process to yield the coloring as in Fig. 11e.

In the remaining square in the second last row, we may use a color which has been used in that row, for instance, the color 8, as in Fig. 12a. In the third last row, the colors $14, 15$ may be used as in Fig. 12b. In the fourth last row, the colors smaller than 22 may be used as in Fig. 12c. One may continue this process to yield the coloring as in Fig. 12d. Note that the coloring as in Fig. 12d does satisfy the required condition. ♣

Now that we have gained some experience, we may proceed to the general case as follows.

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Bogus Solution. Suppose we have colored the n^2 unit squares using r colors so that the required condition is met. Consider the diagonal joining the upper right corner with the lower left corner. Note that the unit squares of the bottom row have distinct colors. Moreover, the unit squares lying to the right of the above-mentioned diagonal and contained in a row except the bottom one, have colors different from the unit squares contained in the rows below and lying to the right of that diagonal. Hence, we require at least

$$n + (n-1) + (n-2) + \cdots + 2 + 1 = \frac{1}{2}n(n+1)$$

colors.

Moreover, similar to the coloring as in Fig. 12, we may color the n^2 unit squares using $\frac{1}{2}n(n+1)$ colors so that required condition is satisfied.

This proves that the least number of colors required is $\frac{1}{2}n(n+1)$.

Question. Where does the above go wrong?

Now that we have gained some idea, we may proceed to the general case as follows.

Solution 18. If $n = 1$, then using one color works. Let us assume[1] that $n \geq 2$.

Suppose we have colored the n^2 unit squares using r colors so that the required condition is met. Consider the diagonal joining the upper right corner with the lower left corner. Note that the unit squares of the bottom row have distinct colors. Moreover, the unit squares lying on the rightmost column except the bottom square, have colors different from those of the squares lying on the bottom row. Hence, we require at least

$$n + (n-1) = 2n - 1$$

colors.

Next, we show that there is a coloring of the n^2 unit squares using $2n-1$ colors satisfying the required condition. Indeed, if the colors $i, i+1, \ldots, i+n-1$ are applied to the squares lying in the i-th last row in an increasing order from the left to the right for all $1 \leq i \leq n$, then the squares in any row have distinct colors and so are the squares in any column, and hence, it is a coloring satisfying the required condition.

This shows that the least number of colors required is $2n-1$. ■

Exercise 22 (RMO 2016c P4,). There are 100 countries participating in an olympiad. Suppose n is a positive integer such that each of the 100 countries is willing to communicate in exactly n languages. If each set of 20 countries can communicate in at least one common language, and no language is common to all 100 countries, what is the minimum possible value of n?

[1] The reason for considering the case $n = 1$ separately will become clear from the rest of the argument. Is it clear what would go wrong with the rest of the argument if $n = 1$?

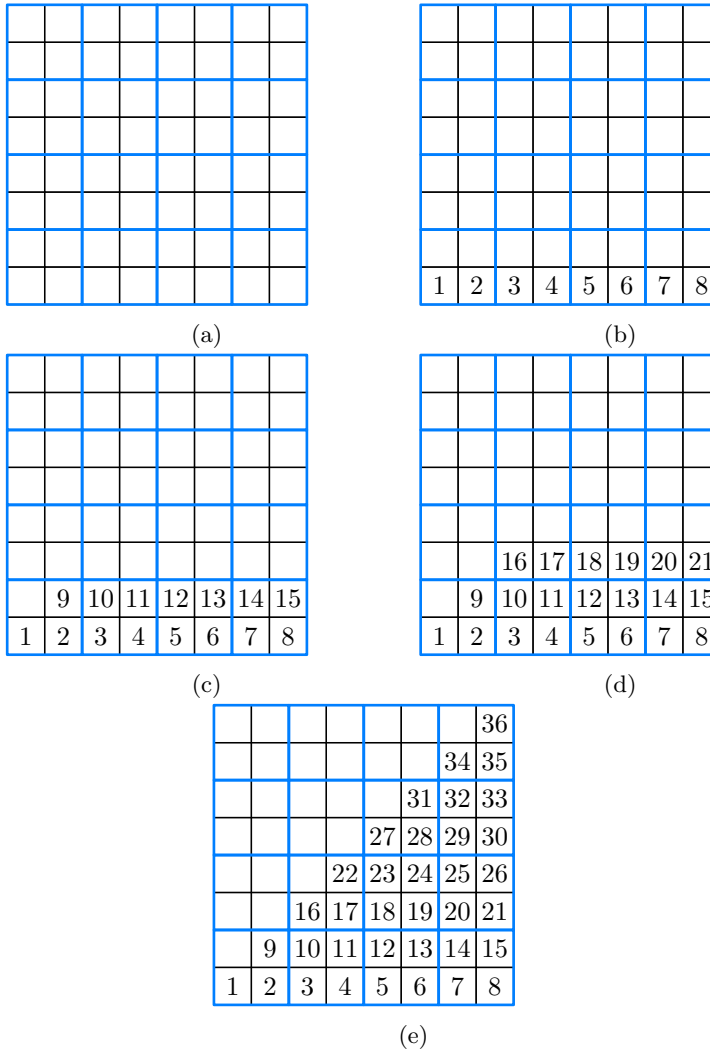


Figure 11: India RMO 2017a P4, Exercise 21

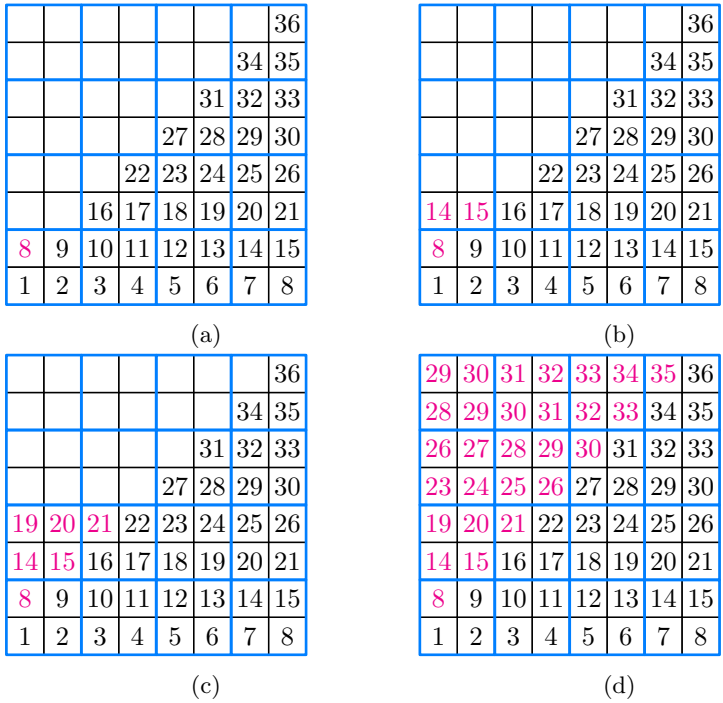


Figure 12: India RMO 2017a P4, Exercise 21

Walkthrough —

(a) Choose a country C, and let L_1 denote a language in which C communicates. Note that L_1 is not common to some country C_1. Hence, some language L_2 is common to the countries C, C_1.

(b) The language L_2 is not common to some country C_2. Hence, some language L_3 is common to the countries C, C_1, C_2.

(c) **Continuing** this line of argument, show that there are pairwise distinct languages $L_1, \ldots, L_{20}$ and C communicates in them.

(d) Conclude that $n \geq 20$.

(e) Show that 20 is the minimum possible value of n.

Solution 19.

Claim — The inequality $n \geq 20$ holds.

Proof of the Claim. Let C denote a country. Let L_1 denote a language in which C communicates. By hypothesis, the language L_1 is not common to some country C_1. Some language L_2 is common to the countries C, C_1. By hypothesis, L_2 is not common to some country C_2. Note that some language L_3 is common to the countries C, C_1, C_2.

Suppose for some integer $2 \leq i$, there are countries $C_1, C_2, \ldots, C_i$ and there are languages $L_1, L_2, \ldots, L_{i+1}$ such that for any $1 \leq k \leq i$, the language L_{k+1} is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and L_k is not common to C_k. By the hypothesis, L_{i+1} is not common to some country C_{i+1}. If $i+2 \leq 20$, then the $i+2$ countries $C, C_1, \ldots, C_{i+1}$ have some common language L_{i+2}. By induction, there are countries $C_1, C_2, \ldots, C_{19}$, and there are languages $L_1, \ldots, L_{20}$ such that for any $1 \leq k \leq 19$, the language L_{k+1} is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and that the language L_k is not common to the country C_k. For any $1 \leq k \leq 19$, the language L_{k+1} is common to the $k+1$ countries $C, C_1, \ldots, C_k$, and for any $1 \leq i \leq k$, the language L_i is not common to C_i, and hence, L_{k+1} is not equal to any of $L_1, \ldots, L_k$. This proves that the languages $L_1, \ldots, L_{20}$ are pairwise distinct, and C communicates in them. The Claim follows. □

Claim — One possible value of n is 20.

Proof of the Claim. Let $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ be pairwise languages. Let $\mathscr{C}_1, \ldots, \mathscr{C}_{100}$ be a choice of 100 countries. Suppose for $1 \leq i \leq 21$, the country $\mathscr{C}_i$ communicates in all the languages $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ other than $\mathscr{L}_i$. Assume further that for any $22 \leq i \leq 100$, the country $\mathscr{C}_i$ communicates in $\mathscr{L}_1, \ldots, \mathscr{L}_{20}$. Note that each country communicates in exactly 20 languages, and that no language is

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common to all the 100 countries. Moreover, each set of 20 countries among $\mathscr{C}_1, \ldots, \mathscr{C}_{21}$ share exactly one common language.

Note that at most 19 countries from $\mathscr{C}_1, \ldots, \mathscr{C}_{21}$, share at least two languages among $\mathscr{L}_1, \ldots, \mathscr{L}_{21}$ in common, and hence share at least one language among $\mathscr{L}_1, \ldots, \mathscr{L}_{20}$ in common. Hence, each set of 20 countries among $\mathscr{C}_1, \ldots, \mathscr{C}_{100}$, having at most 19 countries from $\mathscr{C}_1, \ldots, \mathscr{C}_{20}$, share at least one language in common. This proves the Claim. □

Combining the above Claims, it follows that the minimum possible value of n is 20. ■

Exercise 23 (RMO 2018b P4,). Suppose 100 points in the plane are coloured using two colours, red and white, such that each red point is the centre of a circle passing through at least three white points. What is the least possible number of white points?

Summary — It relies on the fact that one can find enough points on the plane such that no three of them are collinear and no four of them are concyclic.

Walkthrough —

(a) There is an upper bound on the number of the red points in terms of the number of the white points. This gives an upper bound on the total number of points, which is 100, in terms of the number of the white points.

(b) Use this bound to guess the least possible number of the white points, which would turn out to be 10.

(c) Begin with 10 white points on the plane in *general position*, and then, introduce enough red points to construct a configuration of 100 points with the stated properties.

Solution 20. Let n denote the number of white points. Since each red point is the centre of a circle passing through at least three white points, it follows that the number of red points is at most $\binom{n}{3}$. This shows that

$$n + \binom{n}{3} \geq 100.$$

Note that $n \mapsto n + \binom{n}{3}$ defines an increasing function on the nonnegative integers. Observe that

$$9 + \binom{9}{3} = 93, \quad 10 + \binom{10}{3} = 130.$$

This implies that $n \geq 10$.

We claim that there is a configuration of 100 points on the plane such that it admits a coloring using two colors, red and white, such that precisely 10 points are colored white, and that each red point is the centre of a circle passing through at least three white points. Indeed, consider 10 points on the plane such that no three of them are collinear and no four of them are concyclic [2]. Color these 10 points white. These white points have $\binom{10}{3} = 120$ subsets of size 3. Consider only 90 such subsets of the white points, and for any such subset of size 3, color the center of the circle passing through them red. Since no three white points are collinear and no four white points are concyclic, it follows that there are precisely 90 pairwise distinct red points. So, the red and the white points together form a set of 100 points such that each red point is the centre of a circle passing through at least three white points. ■

Exercise 24 (INMO 2025 P5, , proposed by Pranjal Srivastava and Rohan Goyal)**.** Greedy goblin Griphook has a regular 2000-gon, whose every vertex has a single coin. In a move, he chooses a vertex, removes one coin each from the two adjacent vertices, and adds one coin to the chosen vertex, keeping the remaining coin for himself. He can only make such a move if both adjacent vertices have at least one coin. Griphook stops only when he cannot make any more moves. What is the maximum and minimum number of coins that he could have collected?

Walkthrough —

(a) Show that 1998 coins can be collected (see Fig. 13).

(b) Show that the difference of the number of coins at the vertices with odd indices, and the number of coins at the vertices with even indices, does not change modulo 3 after any move.

(c) Conclude that no more than 1998 coins can be collected.

(d) Show that 668 coins can be collected.

(e) Show that at least 668 coins are collected when Griphook stops.

The following is from AoPS, and is due to Rohan Goyal.

Solution 21. Denote the regular 2000-gon by $A_1A_2 \dots A_{2000}$. The claims below imply that the maximum (resp. minimum) number of coins that Griphook could have collected is 1998 (resp. 668).

Claim — No more than 1998 coins can be collected.

Proof of the Claim. Note that after any move, the number of coins at the vertices with odd indices and the number of coins at the vertices with even

[2] Why does such a collection exist? This could be intuitively clear, but can you write down a precise proof? Does induction help?

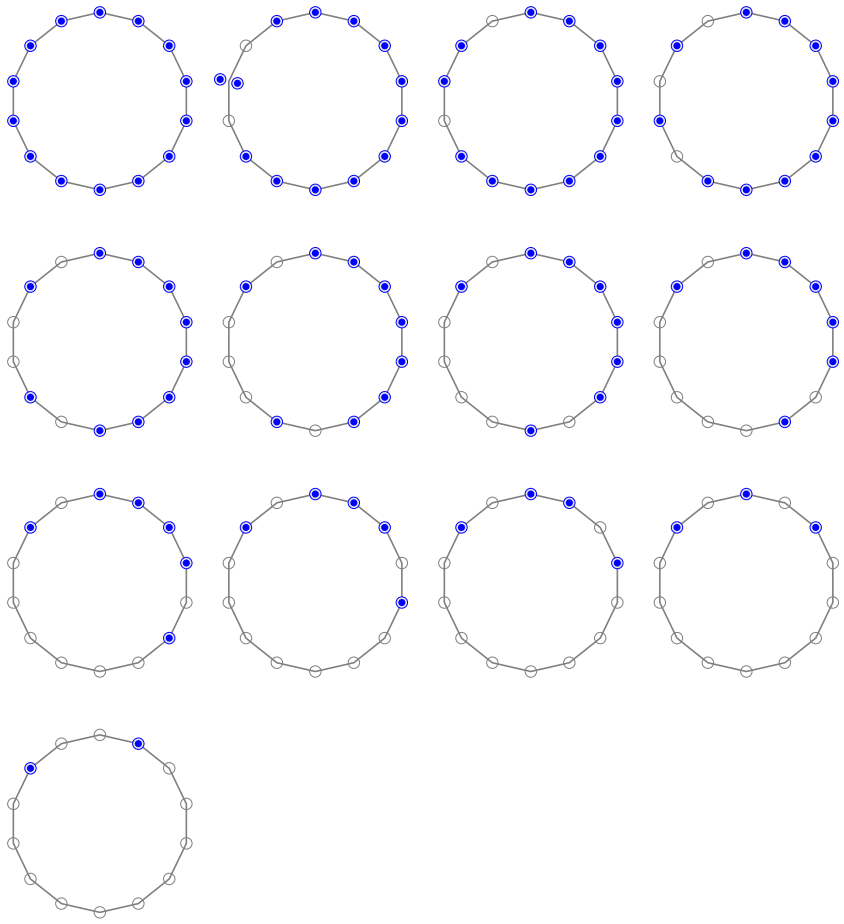


Figure 13: INMO 2025 P5, Collecting 12 coins, Exercise 24

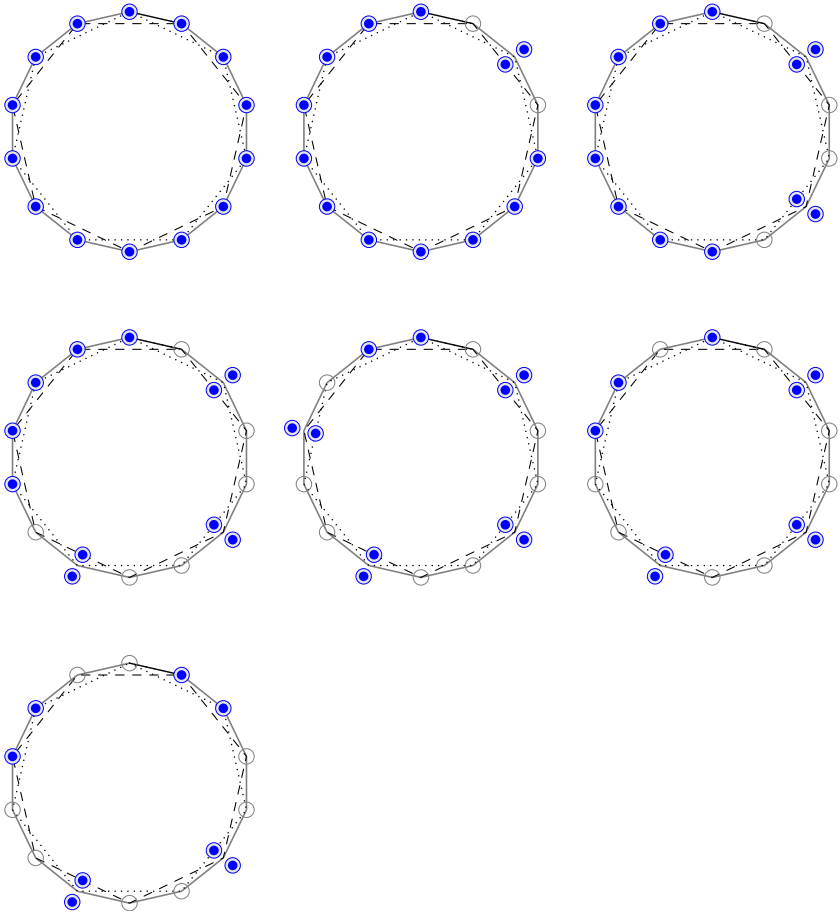


Figure 14: INMO 2025 P5, Collecting 6 coins, Exercise 24

indices either reduces by 2 and increases by 1 respectively, or increases by 1 and reduces by 2 respectively. This shows that the difference of the number of coins at the vertices with odd indices and the number of coins at the vertices with even indices, does not change modulo 3 after any move, and hence, this difference is zero modulo 3 after any move.

Since the number of coins remaining after any move is at least 1, there are at least two coins left at the end of any move. Consequently, no more that 1998 coins can be collected. □

Claim — There is a way to collect 1998 coins.

Proof of the Claim. Let us denote the number of coins at the vertices of the regular 2000-gon by a string of length 2000, consisting of nonnegative integers, where the i-th entry of the string denotes the number of coins at the i-th vertex A_i.

In the following, 1^m (resp. 0^m) denotes a string of m 1's (resp. 0's). Note that by making a move, the string 1^{2000} can be turned to $1^{1996}0201$, which can be turned to $1^{1995}01010$ by making another move. Observe that the sub-string 101 can be turned to 010 by making a move. Hence, making enough suitable moves, the string $1^{1995}01010$ can be reduced to $010^{1996}10$. While reaching $010^{1996}10$ from 1^{2000}, a total of 1998 coins have been earned, proving the claim. □

Claim — There is a sequence of moves, which leads to the collection of 668 coins when Griphook stops.

Proof of the Claim. Write the string 1^{2000} as $1(111)^{666}1$, which can be turned to $1(020)^{666}1$, which is equal to $1020(020)^{664}0201$. This can be reduced to $0110(020)^{664}0110$. Note that no further move is possible, and hence Griphook has to stop, and he earned $2000 - 4 - 2 \times 664 = 668$ coins. □

Claim — For any sequence of moves, Griphook will have at least 668 coins after he stops.

Proof of the Claim. Let c_i denote the number of coins at the i-th vertex. Note that each move reduces the sum

$$S = \sum_{i=1}^{2000} \min\{c_{i-1}, c_{i+1}\}.$$

In this sum, the indices are to be read modulo 2000. In each move, this sum reduces at least by 1, and Griphook stops precisely when this sum reduces to 0.

Note that at any moment, for any two one-vertex apart vertices, one of them contains at most one coin[3], that is,

$$\min\{c_{i-1}, c_{i+1}\} \leq 1$$

for any $1 \leq i \leq 2000$. Write

$$E = \sum_{1 \leq i \leq 2000, 2|i} \min\{c_{i-1}, c_{i+1}\},$$

$$O = \sum_{1 \leq i \leq 2000, 2\nmid i} \min\{c_{i-1}, c_{i+1}\}.$$

Note that in each move, one of E, O is reduced at most by 3. Hence, at least 334 moves are required to reduce any of E, O to 0. Since in each move, only one of E, O gets reduced, it follows that Griphook makes at least $334 + 334 = 668$ moves before he stops.

□

■

Exercise 25 (Moscow Mathematical Olympiad 2025 Grade 9 P4, proposed by A. Zaslavsky)**.** Each cell of a 100×100 square is painted either white or black. It turns out that each white cell has exactly two adjacent cells painted white, and each black cell has exactly two adjacent cells painted black. Find the maximum possible number of black cells.

> **Walkthrough** —
>
> **(a)**

Solution 22.

■

Exercise 26 (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P6, , proposed by S. L. Berlov)**.** Petya chooses 100 pairwise distinct positive numbers less than 1 and arranges them in a circle. In one operation, he may take three consecutive numbers a, b, c (in this order) and replace b with $a - b + c$. What is the greatest value of k such that Petya could initially choose the numbers and perform several operations so that k of the resulting numbers are integers?

[3] Why?

Walkthrough —

(a)

Solution 23. Let $a_1, a_2, \ldots, a_{100}$ be the numbers on the circle in the clockwise order, prior to an operation. Consider the list of numbers, formed by taking the differences of consecutive numbers in the circle:

$$a_1 - a_2, a_2 - a_3, \ldots, a_{100} - a_1.$$

We call this list the *difference list* of the numbers on the circle. Petya chooses a_{i-1}, a_i, a_{i+1} (indices are taken modulo 100) and replaces a_i with $a_{i-1} - a_i + a_{i+1}$. Then the difference list changes as follows:

$$a_{i-1} - a_i, a_i - a_{i+1}$$

is replaced with

$$a_{i-1} - (a_{i-1} - a_i + a_{i+1}) = a_i - a_{i+1}, (a_{i-1} - a_i + a_{i+1}) - a_{i+1} = a_{i-1} - a_i.$$

In other words, the difference list remains unchanged, except that the two numbers $a_{i-1} - a_i$ and $a_i - a_{i+1}$ are swapped. Therefore, the difference list remains invariant up to reordering under the operations.

Since Petya chooses 100 pairwise distinct positive numbers less than 1, the difference list consists of 100 nonzero numbers less than 1 in absolute value. So, after performing several operations, no two adjacent numbers on the circle can be integers, and hence, at most 50 of the numbers on the circle can be integers. This shows that $k \leq 50$.

To show that $k = 50$ is achievable, Petya can initially place the numbers $c_1, c_2, \ldots, c_{100}$ on the circle in the clockwise order, where

$$c_i = \begin{cases} 1 - \frac{1}{2^i} & \text{if } i \text{ is even,} \\ 1 - \frac{1}{2^{i-1}} - \frac{1}{2^{i+1}} & \text{if } i \text{ is odd and } i \neq 1, \\ 1 - \frac{1}{2^{100}} - \frac{1}{2^2} & \text{if } i = 1. \end{cases}$$

Note that operating on c_i for all odd i, leads to having 1 at all odd positions, and hence 50 integers on the circle. This proves that $k = 50$ is achievable.

Hence, the greatest value of k is 50. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)